

IOT ASSIGNMENT 3

Date- 05-07-2026

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Q35. REST API – Weather Data Fetcher

Github link :- https://github.com/ananyabhagat06-ai/summer-internship-2026-iot-/tree/963d4f6dd68506bbcb92e177f7cb85ed9642b3a9/Assignment%20/Q35_Weather_Fetcher

README is available under week 4 folder.

Q36 :-MQTT & HTTP

MQTT (Message Queuing Telemetry Transport) is a lightweight communication protocol mainly designed for IoT devices. It follows the Publish/Subscribe model, where devices send data to a broker, and the broker forwards it only to the devices that have subscribed to that data. MQTT uses very little bandwidth and power, provides low latency, and works well even on slow or unstable internet connections. It also supports security through TLS/SSL encryption, authentication, and access control.

HTTP (Hypertext Transfer Protocol) is the standard protocol used for websites and web applications. It follows the Client/Server model, where a client sends a request and the server returns a response. HTTP consumes more bandwidth and power than MQTT and has higher latency because a new request is required whenever data is needed. It is secured using HTTPS (SSL/TLS) and authentication, making it suitable for web services and REST APIs.

Which protocol is better for a 1000-node Smart Agriculture deployment?

For a smart agriculture system with about 1000 sensor nodes, MQTT is the better choice.

Reasons:

It uses very little bandwidth, making communication efficient.

It consumes less power, which helps battery-operated sensors last longer.

It provides fast, real-time communication with low latency.

A single MQTT broker can manage thousands of connected devices efficiently.

Only devices that subscribe to specific data receive it, reducing unnecessary network traffic.

It performs reliably even on low-bandwidth or unstable network connections.

Conclusion:

MQTT is more suitable than HTTP for large-scale IoT applications like smart agriculture because it is lightweight, energy-efficient, fast, and scalable.

Q37:- WiFi_Security_IoT

ANSWER :- wi-Fi security is essential in IoT because devices such as ESP32 boards often transmit sensitive information over wireless networks.

One common attack vector is WPA2 vulnerabilities, where attackers exploit weaknesses in older security protocols to intercept or manipulate network traffic.

Another common attack is the Evil Twin Attack, in which an attacker creates a fake Wi-Fi access point with the same name as a trusted network.

If an IoT device connects to this fake network, the attacker can monitor communications or steal sensitive data. To secure ESP32-based IoT devices, developers should follow several best practices. Using WPA3 instead of WPA2 provides stronger encryption and better protection against password-guessing attacks. Certificate pinning ensures that the device communicates only with trusted servers by verifying a known server certificate, reducing the risk of man-in-the-middle attacks. Developers should also use secure communication protocols such as HTTPS or MQTTS, keep firmware updated, and use strong, unique Wi-Fi passwords.

Hardcoding Wi-Fi credentials directly into the source code is a poor security practice. If the code is uploaded to a public GitHub repository or accessed by an unauthorized person, the network credentials become exposed. Attackers can use these credentials to gain unauthorized access to the

Wi-Fi network, compromise IoT devices, or launch further attacks. A better approach is to store credentials in a separate configuration file that is excluded from version control using a .gitignore file or to securely provision credentials during device setup.

Q38 :- LoRa_LPWAN

ANSWER :- LPWAN (Low Power Wide Area Network) is a wireless communication technology designed for IoT devices that need to transmit small amounts of data

over long distances while consuming very little power. It is commonly used in applications such as smart agriculture, environmental monitoring, smart cities, and utility metering, where devices often run on batteries for several years.

LoRa (Long Range) achieves long-distance communication using a modulation technique called the Spreading Factor (SF). A higher spreading factor

spreads the signal over a longer period, making it more resistant to interference and allowing it to travel farther. Although higher spreading

factors reduce data transmission speed, they significantly improve communication range while maintaining low power consumption.

FOR FULL ANSWER VISIT GITHUB LINK :-

https://github.com/ananyabhagat06-ai/summer-internship-2026-iot-/blob/963d4f6dd68506bbcb92e177f7cb85ed9642b3a9/Assignment%20/Q38_LoRa_LPWAN

Q39:- IOT GATEWAY

ANSWER :- An IoT Gateway is a device that acts as a bridge between IoT sensor nodes and the cloud. It collects data from multiple sensors, processes or filters it if required, converts data between different communication protocols, and securely forwards the information to cloud platforms. An IoT gateway also improves security, reduces network traffic, and allows communication between devices that use different technologies.

FOR FULL DETAILS CLICK GITHUB LINK

https://github.com/ananyabhagat06-ai/summer-internship-2026-iot-/blob/963d4f6dd68506bbcb92e177f7cb85ed9642b3a9/Assignment%20/IOT_GATEWAY/Q39

Q40:- MQTT_QoS

ANSWER :- Quality of Service (QoS) in MQTT defines the reliability level of message delivery between the publisher and the subscriber. MQTT provides three QoS levels depending on how important the data is and the required reliability.

FULL DETAIL ANSWER LINK

https://github.com/ananyabhagat06-ai/summer-internship-2026-iot-/blob/963d4f6dd68506bbcb92e177f7cb85ed9642b3a9/Assignment%20MQ40_MQTT_QoS/Q40

Module 5: Advanced IoT Projects & Problem Statements

Q41. Smart Home Air Quality Monitor

Github link: <https://github.com/ananyabhagat06-ai/summer-internship-2026-iot-/tree/963d4f6dd68506bbcb92e177f7cb85ed9642b3a9/Assignment%20/Q41/Home%20air%20quality>

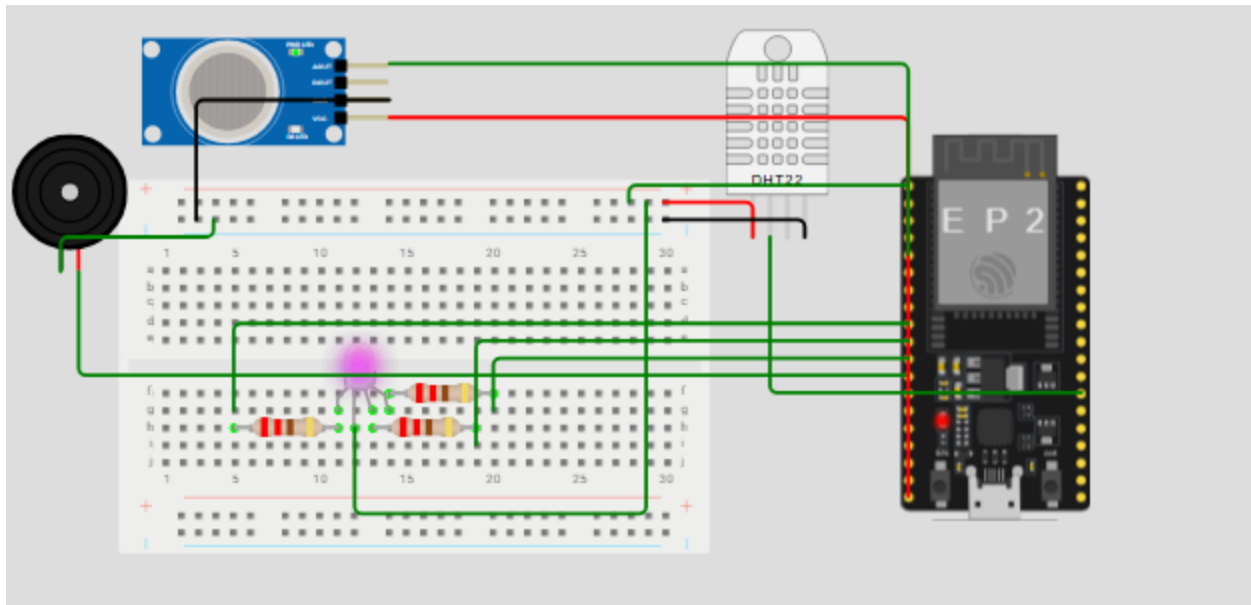
Wokwi Link: <https://wokwi.com/projects/468832410664276993>

The screenshot shows the Wokwi IDE interface for a project titled "smart home air quality". The left pane contains the following code:

```
1 #include <DHT.h>
2
3 // ----- DHT SETUP -----
4 #define DHTPIN 4
5 #define DHTTYPE DHT22
6 DHT dht(DHTPIN, DHTTYPE);
7
8 // ----- MQ2 -----
9 #define MQ2_PIN 34
10
11 // ----- OUTPUT PINS -----
12 #define RED_PIN 25
13 #define GREEN_PIN 26
14 #define BLUE_PIN 27
15 #define BUZZER 14
16
17 int baseline = 0;
18
19 // ----- RGB FUNCTION -----
20 void setColor(bool r, bool g, bool b) {
21   // If LED behaves wrong -> invert (!r !g !b)
22   digitalWrite(RED_PIN, r);
23   digitalWrite(GREEN_PIN, g);
24   digitalWrite(BLUE_PIN, b);
25 }
26
27 // ----- SETUP -----
28 void setup() {
29   Serial.begin(115200);
30   dht.begin();
```

The right pane shows a simulation of the hardware. It includes a breadboard with a DHT22 sensor, an MQ2 sensor, and an ESP2 module. The console window at the bottom displays the following data:

Time	MQ2 ppm	DHT Temp (°C)	DHT Humidity (%)	Status
00:26	3628	24.00	40.00	SAFE
00:28	3628	24.00	40.00	SAFE
00:30	3628	24.00	40.00	SAFE
00:32	3628	24.00	40.00	SAFE
00:34	3628	24.00	40.00	SAFE
00:36	3628	24.00	40.00	SAFE
00:38	3628	24.00	40.00	SAFE

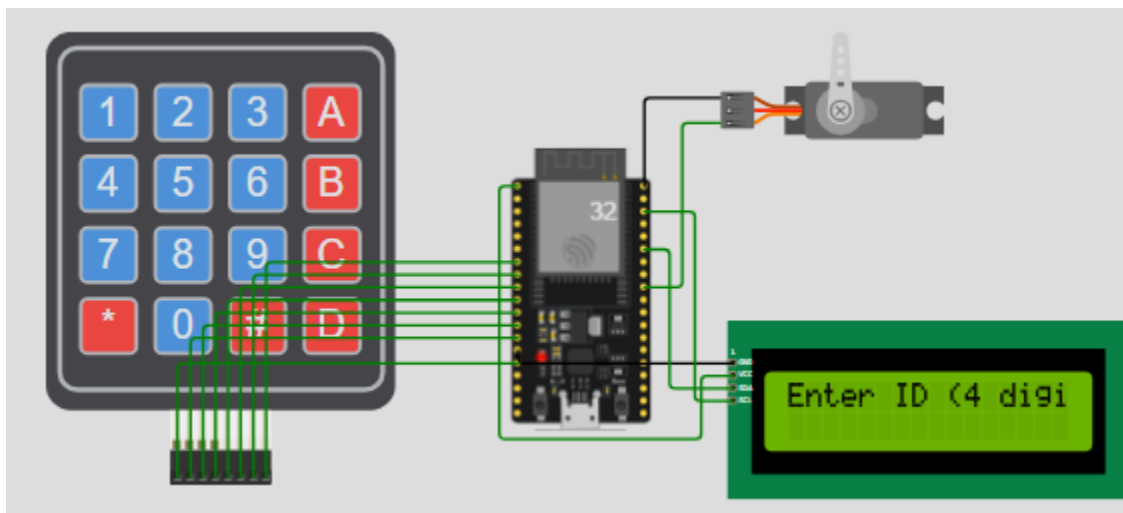


README is available ASSIGNMENT FOLDER (Q41/Home air quality)

Q43. Smart Door Lock with OTP

Github link- <https://github.com/ananyabhagat06-ai/summer-internship-2026-iot-/tree/963d4f6dd68506bbcb92e177f7cb85ed9642b3a9/Assignment%20/Q43>

Wokwi Link- <https://wokwi.com/projects/468834224285625345>

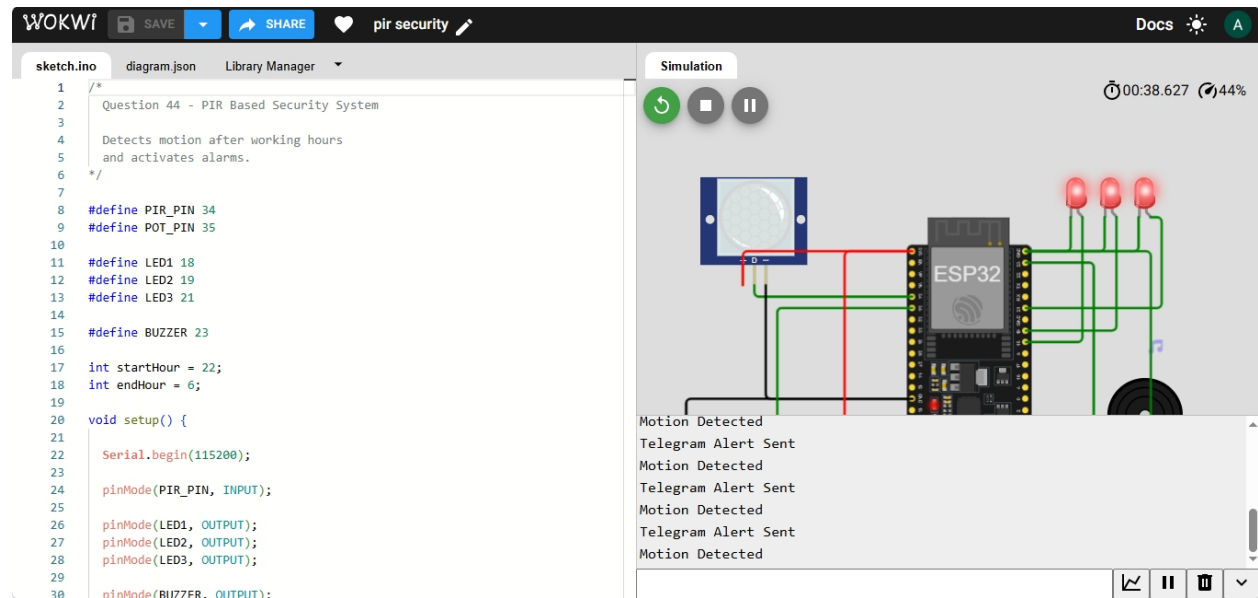


README is under ASSIGNMENT FOLDER (Q43)

Q44. PIR-Based Security Camera Trigger

Github Link- https://github.com/ananyabhagat06-ai/summer-internship-2026-iot-/tree/963d4f6dd68506bbcb92e177f7cb85ed9642b3a9/Assignment%20/Q44_pir_sensor

Wokwi link- <https://wokwi.com/projects/468819312805842945>



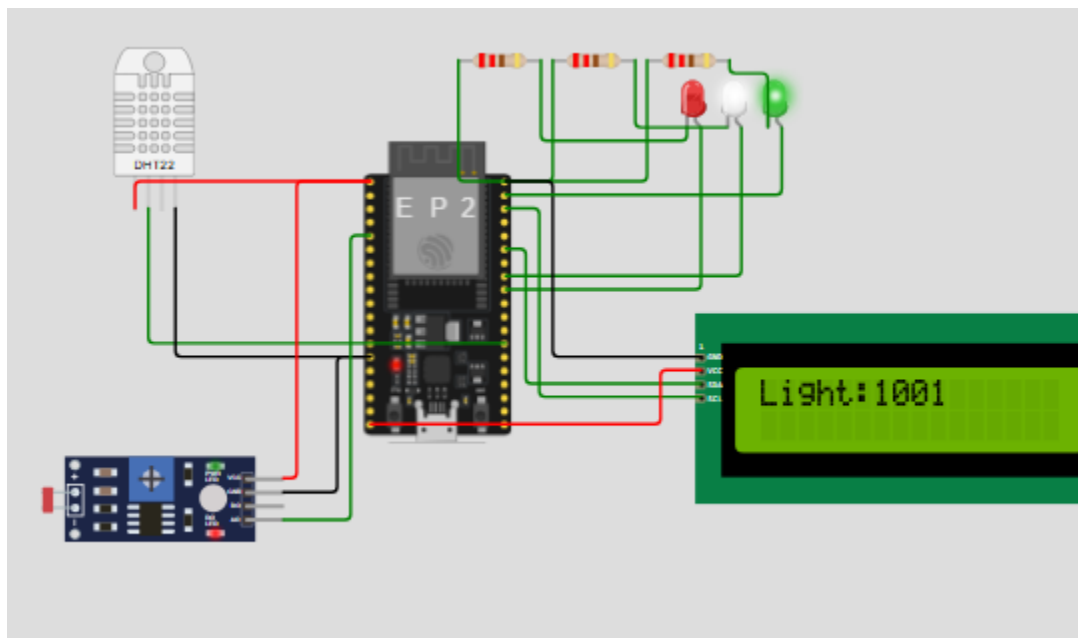
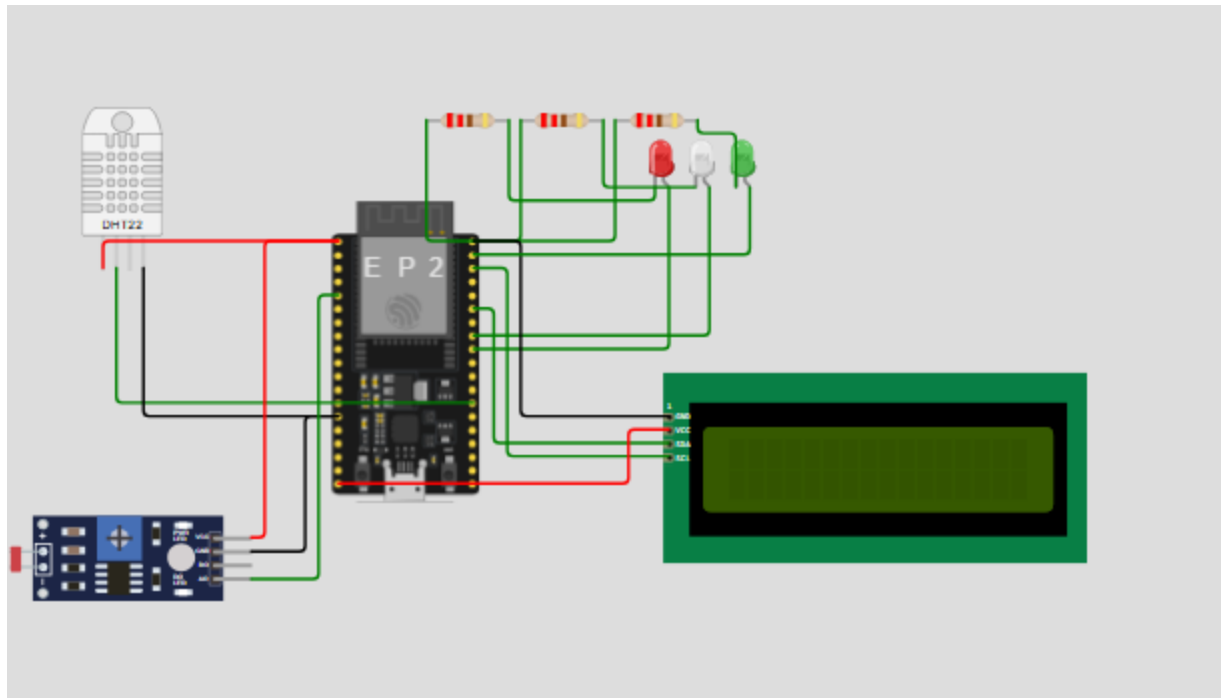
MORE DETAILS CLICK LINK OF GITHUB

UNDER ASSIGNMENT FOLDER (PIR SENSOR)

Q45. Smart Greenhouse Controller

Github :- <https://github.com/ananyabhagat06-ai/summer-internship-2026-iot-/tree/963d4f6dd68506bbcb92e177f7cb85ed9642b3a9/Assignment%20/Q45%20greenhouse>

Wokwi Link- <https://wokwi.com/projects/468821064222597121>



Q46. River Water Level Alert (Flood Detection)

Github link- <https://github.com/ananyabhagat06-ai/summer-internship-2026-iot->

[/tree/963d4f6dd68506bbcb92e177f7cb85ed9642b3a9/Assignment%20/Q46](https://github.com/ananyabhagat06-ai/summer-internship-2026-iot-/tree/963d4f6dd68506bbcb92e177f7cb85ed9642b3a9/Assignment%20/Q46)

Wokwi link- <https://wokwi.com/projects/468822491582194689>

WOKWI

SAVE

SHARE

♥

flood detection

Docs

A

sketch.ino

diagram.json

Library Manager

```

1  /*
2  | Question 46 - River Water Level Alert
3  | Summer School IoT Assignment
4  | */
5
6  #define TRIG_PIN 5
7  #define ECHO_PIN 18
8
9  #define GREEN_LED 19
10 #define YELLOW_LED 21
11 #define RED_LED 22
12
13 #define BUZZER 23
14
15 float samples[10];
16 int indexPos = 0;
17
18 float getDistance() {
19
20     digitalWrite(TRIG_PIN, LOW);
21     delayMicroseconds(2);
22
23     digitalWrite(TRIG_PIN, HIGH);
24     delayMicroseconds(10);
25
26     digitalWrite(TRIG_PIN, LOW);
27
28     long duration = pulseIn(ECHO_PIN, HIGH);
29
30     return duration * 0.034 / 2.0;

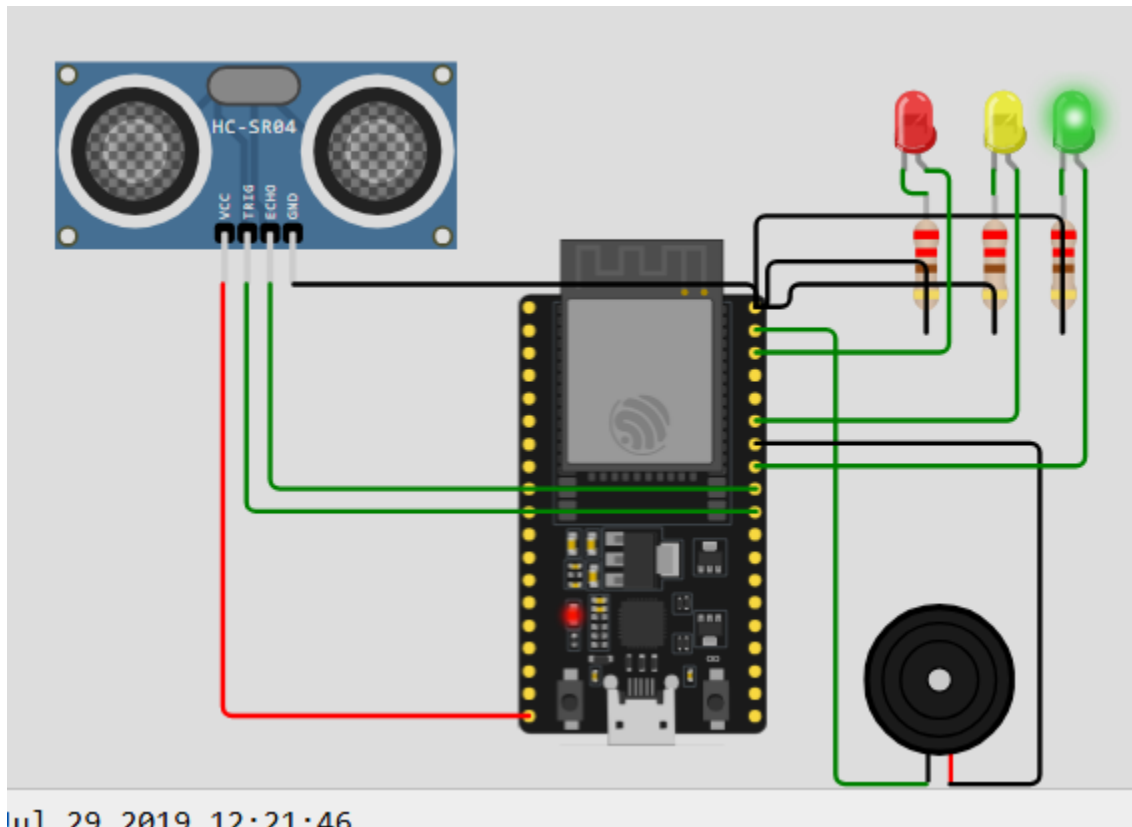
```

Simulation

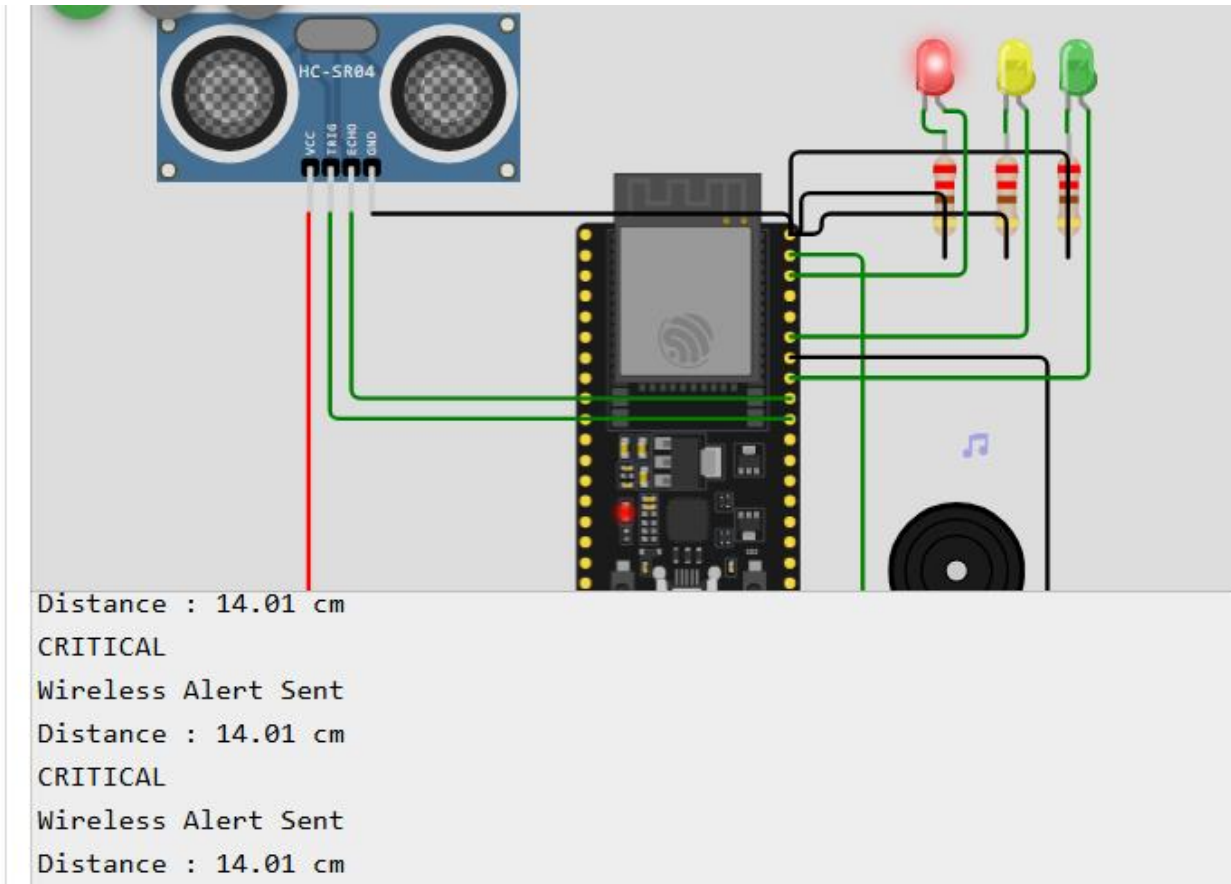
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+

⋮



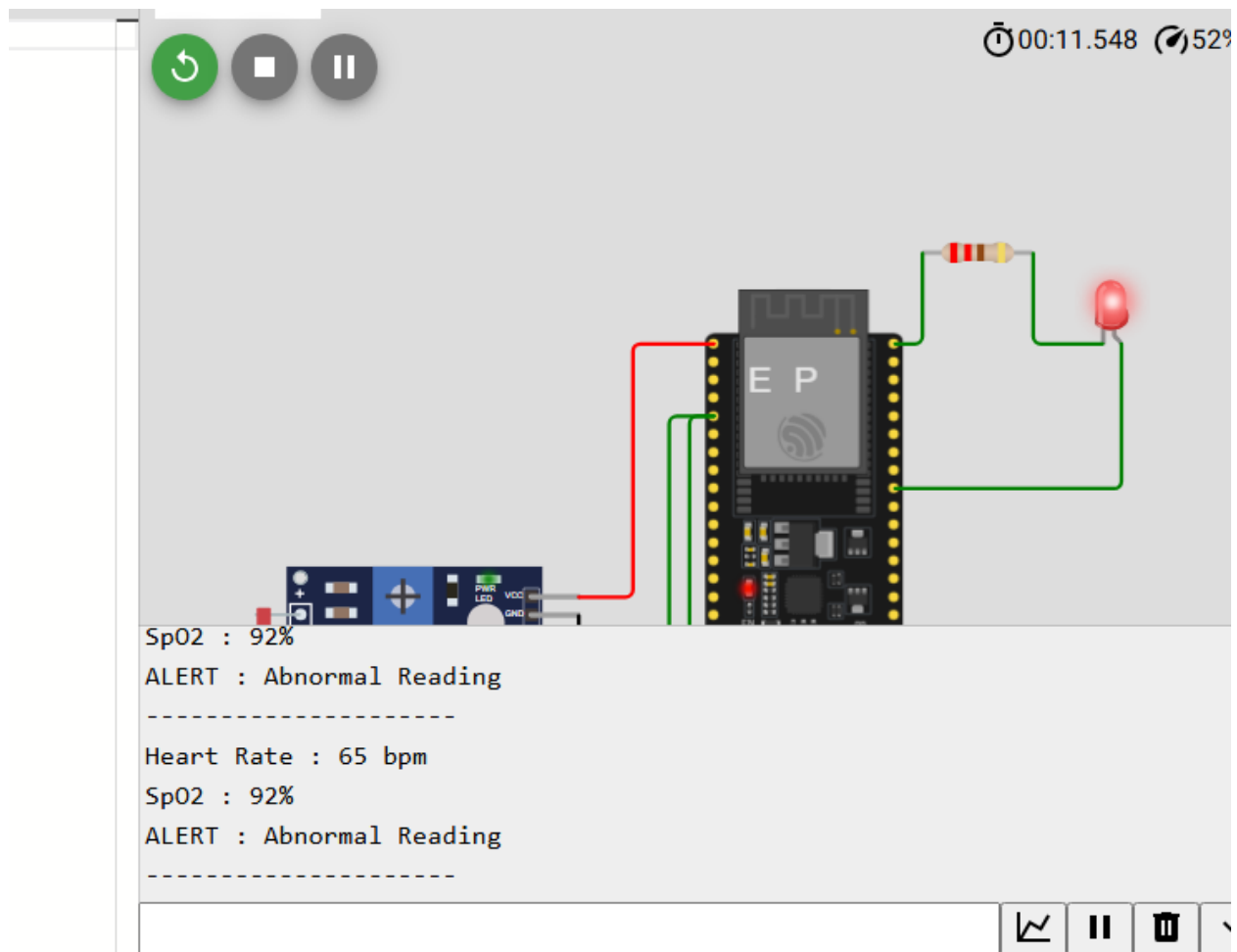
Jul 29 2019 12:21:46



Q47. Heart Rate & SpO2 Monitor (Simulation)

Github link- [https://github.com/ananyabhagat06-ai/summer-internship-2026-iot-
/tree/963d4f6dd68506bbcb92e177f7cb85ed9642b3a9/Assignment%20/Q47
%20_health_monitor](https://github.com/ananyabhagat06-ai/summer-internship-2026-iot-/tree/963d4f6dd68506bbcb92e177f7cb85ed9642b3a9/Assignment%20/Q47%20_health_monitor)

Wokwi Link- <https://wokwi.com/projects/468826261137777665>



Q48. Design Challenge: IoT for Jammu Smart City

Github link- <https://github.com/ananyabhagat06-ai/summer-internship-2026-iot-/tree/ce781d4b42083359566965762aa8d30f10b11098/Final%20project%20>

Q49. Security Vulnerability Analysis

Github link- <https://github.com/ananyabhagat06-ai/summer-internship-2026-iot-/tree/b59fc6ef49fd085b8131c633eda85ece87f761f8/Assignment%20/Q49%20/Security%20Vulnerability%20Analysis%20>

Q50. Reflection & Learning Portfolio

Github link- <https://github.com/ananyabhagat06-ai/summer-internship-2026-iot-/blob/d96a8a074cf664ab0f6728226f19b19be3107380/Assignment%20/Reflection%20%26%20Learning%20Portfolio>

THANK YOU.